

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250270772

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

Brenner; Mark W. et al.

VIBRATOR FREQUENCIES FOR SLIPFORM PAVERS

Abstract

Slipform paver may include rheometers for measuring rheological parameters. The rheometers may include vane rheometers in a grout box of the slipform mold. The slipform paver may dynamically determining the rheological parameters of concrete and controlling a target vibrational frequency of vibrators based on the rheological parameters. The slipform paver may also use depth sensors, cameras, and/or electric vibrators for controlling the target vibration frequency.

Inventors: Brenner; Mark W. (Ida Grove, IA), Farr; Thomas C. (Ida Grove, IA)

Applicant: GOMACO Corporation (Ida Grove, IA)

Family ID: 1000008477569

Appl. No.: 19/048274

Filed: February 07, 2025

Related U.S. Application Data

us-provisional-application US 63557176 20240223

Publication Classification

Int. Cl.: E01C19/48 (20060101); B06B1/18 (20060101); E01C23/01 (20060101); H02K7/06 (20060101)

U.S. Cl.:

CPC E01C19/4893 (20130101); B06B1/186 (20130101); E01C23/01 (20130101); H02K7/061 (20130101); B06B2201/70 (20130101); E01C2301/04 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Application Ser. No. 63/557,176, filed on Feb. 23, 2024, titled “VIBRATOR FREQUENCIES FOR SLIPFORM PAVERS”, which is incorporated herein by reference in the entirety.

TECHNICAL FIELD

[0002] Embodiments of the invention are directed generally toward machines forming and consolidating paving materials, and more particularly to consolidating by vibration.

BACKGROUND OF THE INVENTION

[0003] Slipform pavers use molds with vibrators to consolidate concrete into a slab. Paver operators set the frequency of the vibrators to achieve a desired texture and strength. Setting the frequency too high or too low may cause vibrator trails or strength defects, respectively, in the slab. Therefore, it would be advantageous to provide a device, system, and method that cures the shortcomings described above.

SUMMARY

[0004] In some aspects, the techniques described herein relate to a paving machine including: a slipform mold, the slipform mold configured to form concrete into a slab; the slipform mold including: a plurality of hydraulic vibrators; wherein the plurality of hydraulic vibrators are configured to vibrate in response to receiving hydraulic fluid; the plurality of hydraulic vibrators including a plurality of vibration sensors configured to measure actual vibrations of the plurality of hydraulic vibrators; a finishing pan; a pair of side plates; and one or more rheometers; wherein the one or more rheometers are configured to measure one or more rheological parameters of the concrete; a hydraulic power supply including a hydraulic pump; a hydraulic manifold; wherein the hydraulic manifold is coupled between the hydraulic pump and the plurality of hydraulic vibrators; and a controller including one or more processors configured, via executable code, to: cause the hydraulic manifold to control an actual flow rate of the hydraulic fluid from the hydraulic pump to the plurality of hydraulic vibrators towards target frequencies using the actual frequencies as feedback; and dynamically update the target frequencies based on the one or more rheological parameters.

[0005] It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not necessarily restrictive of the present disclosure. The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate subject matter of the disclosure. Together, the description and drawings serve to explain the principles of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] The numerous advantages of the disclosure may be better understood by those skilled in the art by reference to the accompanying figures in which:

[0007] FIGS. 1A-1B depicts a front perspective view of a paving machine, in accordance with one or more embodiments of the present disclosure.

[0008] FIG. 1C depicts a side view of a slipform mold including a rheometer, in accordance with one or more embodiments of the present disclosure.

[0009] FIG. 1D depicts a front view of a paving machine, in accordance with one or more embodiments of the present disclosure.

[0010] FIG. 1E depicts a front perspective view of a slipform mold, in accordance with one or

more embodiments of the present disclosure.

[0011] FIG. 1F depicts a side view of a slipform mold including a rheometer, in accordance with one or more embodiments of the present disclosure.

[0012] FIG. 1G depicts a simplified circuit diagram of a paving machine using rheological parameters from a rheometer to dynamically update target frequencies of hydraulic vibrators, in accordance with one or more embodiments of the present disclosure.

[0013] FIG. 2A depicts a front perspective view of a paving machine with depth sensors, in accordance with one or more embodiments of the present disclosure.

[0014] FIG. 2B depicts a simplified circuit diagram of a paving machine using depth maps from depth sensors to dynamically update target frequencies of hydraulic vibrators, in accordance with one or more embodiments of the present disclosure.

[0015] FIG. 3A depicts a side view of a paving machine including cameras, in accordance with one or more embodiments of the present disclosure.

[0016] FIG. 3B depicts a rear perspective view of a paving machine including cameras, in accordance with one or more embodiments of the present disclosure.

[0017] FIG. 3C depicts a simplified circuit diagram of a paving machine using images from cameras to dynamically update target frequencies of hydraulic vibrators, in accordance with one or more embodiments of the present disclosure.

[0018] FIG. 3D depicts an example image with surface voids which are relatively large, in accordance with one or more embodiments of the present disclosure.

[0019] FIG. 3E depicts an example image with surface voids which are relatively small, in accordance with one or more embodiments of the present disclosure.

[0020] FIG. 4A depicts a front perspective view of a paving machine with electric vibrators, in accordance with one or more embodiments of the present disclosure.

[0021] FIG. 4B depicts a simplified circuit diagram of a paving machine with electric vibrators, in accordance with one or more embodiments of the present disclosure.

[0022] FIG. 5A depicts a front perspective view of a paving machine with an electric vibrator which acts as a sensor in a bank of hydraulic vibrators, in accordance with one or more embodiments of the present disclosure.

[0023] FIG. 5B depicts a simplified circuit diagram of a paving machine with an electric vibrator which acts as a sensor in a bank of hydraulic vibrators, in accordance with one or more embodiments of the present disclosure.

[0024] FIGS. 6A-6B depict a top view of a system including a paving machine and a mixer truck, in accordance with one or more embodiments of the present disclosure.

[0025] FIG. 6C depicts a simplified circuit diagram of a paving machine with meteorological sensors which are used to update rheological parameters received from the mixer truck due to water evaporation over time, in accordance with one or more embodiments of the present disclosure.

[0026] FIG. 6D depicts an example graph of a computation performed by a controller when updating the rheological parameter over time using the meteorological data for compensation where the rheological parameter is slump and the meteorological data is temperature, in accordance with one or more embodiments of the present disclosure.

[0027] FIG. 7 depicts a front perspective view of the paving machine configured as a two-track machine, in accordance with one or more embodiments of the present disclosure.

[0028] FIG. 8A depicts a rear perspective view of a paving machine configured as a curb-and-gutter paving machine, in accordance with one or more embodiments of the present disclosure.

[0029] FIG. 8B depicts a front perspective view of a paving machine configured as a barrier paving machine, in accordance with one or more embodiments of the present disclosure.

[0030] FIG. 8C depicts a partial perspective view inside a hopper of an offset slipform mold of the paving machine, in accordance with one or more embodiments of the present disclosure.

DETAILED DESCRIPTION OF THE INVENTION

[0031] Before explaining one or more embodiments of the disclosure in detail, it is to be understood that the embodiments are not limited in their application to the details of construction and the arrangement of the components or steps or methodologies set forth in the following description or illustrated in the drawings. In the following detailed description of embodiments, numerous specific details may be set forth in order to provide a more thorough understanding of the disclosure. However, it will be apparent to one of ordinary skill in the art having the benefit of the instant disclosure that the embodiments disclosed herein may be practiced without some of these specific details. In other instances, well-known features may not be described in detail to avoid unnecessarily complicating the instant disclosure.

[0032] As used herein a letter following a reference numeral is intended to reference an embodiment of the feature or element that may be similar, but not necessarily identical, to a previously described element or feature bearing the same reference numeral (e.g., **1**, **1a**, **1b**). Such shorthand notations are used for purposes of convenience only and should not be construed to limit the disclosure in any way unless expressly stated to the contrary.

[0033] Further, unless expressly stated to the contrary, “or” refers to an inclusive or and not to an exclusive or. For example, a condition A or B is satisfied by any one of the following: A is true (or present) and B is false (or not present), A is false (or not present) and B is true (or present), and both A and B are true (or present).

[0034] In addition, use of “a” or “an” may be employed to describe elements and components of embodiments disclosed herein. This is done merely for convenience and “a” and “an” are intended to include “one” or “at least one,” and the singular also includes the plural unless it is obvious that it is meant otherwise.

[0035] Finally, as used herein any reference to “one embodiment”, “in embodiments”, or “some embodiments” means that a particular element, feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment disclosed herein. The appearances of the phrase “in some embodiments” in various places in the specification are not necessarily all referring to the same embodiment, and embodiments may include one or more of the features expressly described or inherently present herein, or any combination or sub-combination of two or more such features, along with any other features which may not necessarily be expressly described or inherently present in the instant disclosure.

[0036] Reference will now be made in detail to the subject matter disclosed, which is illustrated in the accompanying drawings. Embodiments of the present disclosure are directed to vibrator frequencies for slipform pavers. Slipform paver may include rheometers for measuring rheological parameters. The rheometers may include vane rheometers in a grout box of the slipform mold. The slipform paver may dynamically determining the rheological parameters of concrete and controlling a target vibrational frequency of vibrators based on the rheological parameters. The slipform paver may also use depth sensors, cameras, and/or electric vibrators for controlling the target vibration frequency.

[0037] Referring generally to FIGS. **1A-1G**, a paving machine **100** is described, in accordance with one or more embodiments of the present disclosure. The paving machine **100**, may be any suitable paving machine, such as, but not limited to, a slipform paver. The paving machine **100** may include one or more components, such as, but not limited to, a slipform mold **102**, frame **104**, end structures **106**, controller **108**, hydraulic power supply **110**, pivot arms **116**, and the like.

[0038] The paving machine **100** may include the slipform mold **102**. The slipform mold **102** may be coupled to the frame **104**. The slipform mold **102** may be an inset mold and mounted below the frame **104**. The slipform mold **102** may be moved in a paving direction for forming concrete **101** into a slab **103**. The concrete **101** may be poured ahead of the slipform mold **102**. For example, the concrete **101** may be poured ahead of the slipform mold **102** by a placer/spreader machine or the like. The slipform mold **102** may consolidate the concrete **101** to form the slab **103**. Thus, the slipform mold **102** may perform slipform paving operations.

[0039] The slipform mold **102** may include one or more components, such as, but not limited to, hydraulic vibrators **118**, grout box auger **120**, tamper bar (not depicted), finishing pan **122**, side plates **124**, split front auger **126**, strike-off **128**, rheometers **130**, and the like.

[0040] The slipform mold **102** may include grout box auger **120**. The grout box auger **120** may also be referred to as a spreader/auger. The grout box auger **120** may span between the pair of the side plates **124**. The grout box auger **120** may be disposed behind and/or above the hydraulic vibrators **118** in the paving direction. The grout box auger **120** may be longitudinally disposed between the hydraulic vibrators **118** and the finishing pan **122**. The grout box auger **120** may spread the concrete **101** across a lateral width of the slipform mold **102** (i.e., perpendicular to the paving direction). The grout box auger **120** may include any suitable screw conveyor design for spreading the concrete **101** across the lateral width. For example, the grout box auger **120** may include a range of suitable pitches, flights, and diameters for spreading the concrete **101**. The grout box auger **120** may be further coupled with a motor, for rotating the grout box auger **120**. A rotational speed of the grout box auger **120** may be adjusted. The rotational speed of the grout box auger **120** may be adjusted based on a speed of the paving machine **100**, the rheological parameters **105**, and the like.

[0041] The slipform mold **102** may include finishing pan **122**. The finishing pan **122** may span between the side plates **124**. The finishing pan **122** may be configured to finish the concrete **101** into the slab **103**. For example, the finishing pan **122** may form the top surface of the slab **103**.

[0042] The slipform mold **102** may include side plates **124**. The side plates **124** may also be referred to as side forms, or the like. The volume between the side plates **124** may be referred to as a grout box. The concrete **101** may be maintained between the side plates **124** as the paving machine **100** travels in the paving direction. The side plates **124** may form the sides of the slab **103** from the concrete **101**.

[0043] The slipform mold **102** may include the split front auger **126** and/or the strike-off **128**. The split front auger **126** may be disposed in front of the strike-off **128** in the paving direction. The split front auger **126** may spread the concrete **101** across a lateral width of the slipform mold **102** (i.e., perpendicular to the paving direction). The split front auger **126** may be split into left-hand and right-hand sections. The left-hand and right-hand sections of the split front auger **126** may be independently driven. The strike-off **128** may span between the side plates **124**. The strike-off **128** may be longitudinally disposed between the split front auger **126** and the hydraulic vibrators **118**. The strike-off **128** may include a flat plate which may strike off the concrete **101** at a select height. Although the slipform mold **102** is described as including the split front auger **126** and the strike-off **128**, this is not intended as a limitation of the present disclosure. The slipform mold **102** may include an open front configuration which may not include the split front auger **126** and the strike-off **128**.

[0044] The slipform mold **102** may include hydraulic vibrators **118**. The hydraulic vibrators **118** may vibrate the concrete **101** for consolidating air out of the concrete **101**. The hydraulic vibrators **118** may be laterally disposed between the side plates **124**. The paving machine **100** may include any number of the hydraulic vibrators **118-1** through **118-n**, where n is an integer. For example, n may include but is not limited to two vibrators, four vibrators, six vibrators, eight vibrators, sixteen vibrators, thirty-two vibrators, or more. The hydraulic vibrators **118** may include any type of vibrator, such as, but not limited to, immersion vibrators (also known as needle vibrators, bent pipe vibrator, internal vibrators), external vibrators (also known as shutter vibrator or formwork vibrator), or surface vibrators (also known as pan vibrators). In embodiments, the hydraulic vibrators **118** are immersion vibrators. For example, the hydraulic vibrators **118** may be vibrators described in U.S. Pat. No. 6,055,486, titled ACCELEROMETER-BASED MONITORING AND CONTROL OF CONCRETE CONSOLIDATION, which is incorporated herein by reference in the entirety.

[0045] The paving machine **100** may include the hydraulic power supply **110**. The hydraulic power

supply **110** may be connected to the frame **104**. The hydraulic power supply **110** may include several components, such as, but not limited to, power sources, hydraulic pumps **136**, hydraulic reservoirs **138**, filters, coolers, heaters, and the like. The power sources may include any power source configured to generate power known in the art, such as, but not limited to, a gasoline engine, a diesel engine, or an electric power source of various sizes and power ratings. The power sources may power the hydraulic pumps **136**. The hydraulic pump **136** may receive a mechanical drive or electrical power from the power source and generates a flow of hydraulic fluid with hydraulic power. The hydraulic reservoir **138** may also be referred to as a hydraulic fluid reservoir. The hydraulic pumps **136** may be connected to the hydraulic reservoir **138** and configured to supply a flow of hydraulic fluid from the hydraulic reservoir **138**. The hydraulic pumps **136** may be configured to pump hydraulic fluid from the hydraulic reservoir **138** and supply a flow (e.g., hydraulic power) of the hydraulic fluid to one or more hydraulic components of the paving machine **100**.

[0046] The hydraulic vibrators **118** may include a hydraulic motor **132**, an eccentric weight **134**, and the like. The hydraulic motor **132** may be fluidically coupled to the hydraulic pumps **136**. The hydraulic pumps **136** may supply a flow of hydraulic fluid to the hydraulic motor **132**. The hydraulic motor **132** may cause the eccentric weight **134** to vibrate in response to receiving the flow of hydraulic fluid. The hydraulic motor **132** may receive actual flow rates **156** (Q.sub.ACTUAL) of the hydraulic fluid. The hydraulic motor **132** may vibrate the eccentric weight **134** at actual frequencies **146** (f.sub.ACTUAL) in response to receiving the actual flow rates **156**. The actual frequencies **146** may be proportional to the actual flow rates **156**. Increases in the actual flow rates **156** of hydraulic fluid will, in turn, create an increase in the actual frequencies **146**. For example, the actual frequencies **146** may scale (e.g., linearly scale) with the actual flow rates **156**. Each of the hydraulic vibrators **118-1** through **118-n** may include a respective of the hydraulic motors **132-1** through **132-n**, the eccentric weights **134-1** through **134-n**, and the like. Similarly, each of the hydraulic vibrators **118-1** through **118-n** may receive a respective of the actual flow rates **156-1** through **156-n** and may vibrate at respective of the actual frequencies **146-1** through **146-n**.

[0047] The paving machine **100** may include a hydraulic manifold **140**. The hydraulic manifold **140** may be coupled between the hydraulic pump **136** and the hydraulic vibrators **118**. The hydraulic manifold **140** may control the actual flow rates **156** of hydraulic fluid to the hydraulic vibrators **118**. For example, the hydraulic manifold **140** may be configured to control the actual flow rates **156** of the hydraulic fluid to hydraulic motors **132**. In embodiments, the hydraulic manifold **140** may include adjustable flow control valves **142**. The hydraulic manifold **140** may include adjustable flow control valves **142-1** through **142-n** for each of the respective of the hydraulic vibrators **118-1** through **118-n**. The adjustable flow control valves **142** may control the actual flow rates **156** of hydraulic fluid from the hydraulic pump **136** to the hydraulic vibrators **118**. The adjustable flow control valves **142** may receive electrical signals from the controller **108**, causing the hydraulic manifold **140** to control the actual flow rates **156** to each of the hydraulic motors **132**. The actual frequencies **146** at which the hydraulic vibrators **118** vibrate may be based on the actual flow rates **156** of hydraulic fluid. Thus, the actual flow rates **156** of hydraulic fluid may be controlled to control the actual frequencies **146** of the hydraulic vibrators **118**. The hydraulic vibrators **118** may include a range of the actual frequencies **146** between which the hydraulic vibrators **118** may be controlled. For example, the actual frequencies **146** of the hydraulic vibrators **118** may be selectively controllable between zero and ten-thousand vibrations per minute (VPMs), or more. The actual frequencies **146** of the hydraulic vibrators **118** may be controlled by varying the actual flow rates **156** of hydraulic fluid to the hydraulic vibrators **118**. For example, the adjustable flow control valves **142-1** through **142-n** may control the actual frequencies **146-1** through **146-n** of the hydraulic vibrators **118-1** through **118-n** by varying the actual flow rates **156-1** through **156-n** of hydraulic fluid.

[0048] The paving machine **100** may include the controller **108**. The controller **108** may be configured to control the hydraulic manifold **140**. For example, the controller **108** may control the adjustable flow control valves **142** of the hydraulic manifold **140**. The controller **108** may control the adjustable flow control valves **142** using one or more electrical signals, for selectively controlling the actual flow rates **156** of the hydraulic fluid to the hydraulic motors **132**, thereby controlling the actual frequencies **146** of the hydraulic vibrators **118**. The controller **108** may independently control the adjustable flow control valves **142**, such that the actual frequencies **146** of the hydraulic vibrators **118** may be independently controlled.

[0049] In embodiments, the hydraulic vibrators **118** may include vibration sensors **144**. The vibration sensors **144** may include, but are not limited to, accelerometers or the like. The vibration sensors **144** may be configured to measure actual frequencies **146** (f.sub.ACTUAL) of the hydraulic vibrators **118**.

[0050] In embodiments, the paving machine **100** may include flow meters **154**. The flow meters **154** may be coupled between the hydraulic manifold **140** and the hydraulic vibrators **118**. The flow meters **154** may measure the actual flow rates **156**. For example, the flow meters **154** may include flow meters **154-1** through **154-n** coupled between respective of the adjustable flow control valves **142-1** through **142-n** and the hydraulic vibrators **118-1** through **118-n**. The flow meters **154-1** through **154-n** may measure the actual flow rates **156-1** through **156-n**. The controller **108** may receive the actual flow rates **156** from the flow meters **154** and/or the actual frequencies **146** of the hydraulic vibrators **118** from the vibration sensors **144** (e.g., vibration sensors **144-1** through **144-n**)

[0051] The controller **108** may use the actual flow rates **156** and/or the actual frequencies **146** as feedback when causing the hydraulic manifold **140** to control the actual flow rates **156** of the hydraulic fluid from the hydraulic pump **136** to the hydraulic vibrators **118** thereby controlling the actual frequencies **146** of the hydraulic vibrators **118**. The controller **108** may use the actual flow rates **156** and/or the actual frequencies **146** as feedback when adjusting the actual flow rates **156** of hydraulic fluid to the hydraulic vibrators **118**, thereby moving the actual flow rates **156** towards the target flow rates **158**. Similarly, the controller **108** may use the actual flow rates **156** and/or the actual frequencies **146** as feedback when adjusting the actual flow rates **156** of hydraulic fluid to the hydraulic vibrators **118**, thereby moving the actual frequencies **146** towards the target frequencies **148**. Either of the actual flow rates **156** and/or the actual frequencies **146** may be used as feedback. The actual flow rates **156** may be easier or more accurate to measure and may provide an indicate of the actual frequencies **146**.

[0052] The target flow rates **158** and/or target frequencies **148** may be preset. The target flow rates **158** and/or target frequencies **148** may be preset in the memory. The controller **108** may include one or more user interfaces (not depicted). The user interfaces may set the target flow rates **158** and/or target frequencies **148**. For example, the controller **108** may include dials, switches, touchscreen controls, or the like for setting the target flow rates **158** and/or target frequencies **148**. A paver operator may input the target flow rates **158** and/or target frequencies **148** using the user interfaces. In this regard, the target flow rates **158** and/or target frequencies **148** may be preset by the human operator.

[0053] The hydraulic vibrators **118** may consolidate the concrete **101**. The hydraulic vibrators **118** may vibrate at the actual frequencies **146** to consolidate the concrete **101**. Consolidating the concrete **101** may remove voids in and/or reduce an air content of the concrete **101**. The actual frequencies **146** of the hydraulic vibrators **118** may control a texture and/or strength of the slab **103**. For example, the consolidation may be proportional to the actual frequencies **146**. Vibrating the concrete **101** at an excessive frequency may cause too much air to escape, thereby causing vibrator trails. Vibrating the concrete **101** at an insufficient frequency may cause too much air to be entrained, thereby reducing a strength of the slab **103**.

[0054] The hydraulic vibrators **118** may be maintained at actual frequencies **146** between the

insufficient frequency and the excessive frequency to achieve the desired texture and/or strength. The insufficient frequency may occur when the hydraulic vibrators **118** receive an insufficient flow rate of the hydraulic fluid. The controller **108** may cause the hydraulic manifold **140** to control the actual flow rates **156** to the hydraulic vibrators **118** towards target frequencies **148** using the actual frequencies **146** as feedback. Similarly, the excessive frequency may occur when the hydraulic vibrators **118** receive an excessive flow rate of the hydraulic fluid. The actual flow rates **156** may be maintained between the insufficient flow rate and the excessive flow rate to maintain the actual frequencies **146** between the insufficient frequency and the excessive frequency. The controller **108** may cause the hydraulic manifold **140** to control the actual flow rates **156** to the hydraulic vibrators **118** towards the target flow rates **158** using the actual flow rates **156** as feedback.

[0055] The concrete **101** may include a composition of cements, water, aggregates (e.g., sand, coarse aggregates), admixtures (e.g., plasticizers, accelerators, retarders, air entrainers, corrosion inhibitors), and the like. The composition of the concrete **101** may be based on a mix design and may vary between batches, mixer trucks, and concrete plants, such that the specific composition of the concrete is not intended to be limiting. The concrete **101** may include rheological parameters **105**. The composition of the concrete **101** may control the rheological parameters **105**. The rheological parameters **105** may include, but are not limited to, workability, yield stress ($\tau_{sub.0}$), plastic viscosity (μ), slump, and the like. The workability may be the ease with which the concrete **101** may be mixed, placed, consolidated, and finished to a homogenous condition. The yield stress may be the minimum stress to initiate or maintain flow. The plastic viscosity may be the resistance to flow once the yield stress is exceeded. The plastic viscosity (μ) may be determined based on shear stress (τ), the yield stress ($\tau_{sub.0}$), and a shear rate (γ) according to the Bingham model (i.e., $\tau = \tau_{sub.0} + \mu \cdot \gamma$). The shear stress (τ) may be the component of force across a cross section of the concrete. The shear rate (γ) the rate of shearing deformation. The slump may refer to the measure of consistency of the concrete **101** equal to the subsidence after removal from a slump cone. The slump may be related to the yield stress and the plastic viscosity. For example, higher yield stresses and higher plastic viscosities may correspond to lower slumps. Concrete with a higher slump may more easily flow than concrete with a lower slump.

[0056] The rheological parameters **105** of the concrete **101** may impact the insufficient frequency and the excessive frequency at which the concrete **101** is vibrated, and similarly the insufficient flow rate and the excessive flow rate of hydraulic fluid to the hydraulic vibrators **118**. The rheological parameters **105** of the concrete **101** may be dynamic and change as the paving machine **100** paves the concrete **101**. For example, the rheological parameters **105** may change due to slump loss (e.g., due to a reduction in water content over time, false setting, etc.). By way of another example, the concrete may not be homogenous along the length of the pour. By way of another example, mixer trucks may source the concrete with different compositions (e.g., from different batch plants). By way of another example, a depth of the concrete may change the rheological parameters **105**. The dynamic nature of the concrete **101** may cause the insufficient frequency and the excessive frequency to change over time, such that fixing the actual frequencies **146** of the hydraulic vibrators **118** and/or the actual flow rates **156** to the hydraulic vibrators **118** at set values may cause spots in the slab **103** with vibrator trails due to over-consolidation and/or spots in the slab **103** which are weakened due to under-consolidation.

[0057] The slipform mold **102** may include rheometers **130**. For example, the rheometers **130** may be laterally disposed between the side plates **124** (e.g., within the grout box). The rheometers **130** may be below and/or behind the grout box auger **120** in the paving direction. The rheometers **130** may be above and/or in front of the finishing pan **122** in the paving direction. For example, the rheometers **130** may be disposed above a bottom surface of the finishing pan **122** such that the rheometers **130** are disposed above the top surface of the slab **103**. Thus, the rheometers **130** may be longitudinally disposed between the hydraulic vibrators **118** and the finishing pan **122** and/or between the grout box auger **120** and the finishing pan **122**. The rheometers **130** may be secured in

a fixed position relative to the side plates **124**. For example, the rheometers **130** may be secured with a minimum distance to the side plates **124**. The minimum distance may be based on a size of the aggregates in the concrete **101**.

[0058] The rheometers **130** may include, but are not limited to, vane rheometers, slump sensors, viscometers (e.g., an in-line viscometer, quartz viscometer, and the like), strain gauges, and the like. In embodiments, the rheometers **130** are vane rheometers. For example, the rheometers **130** may include one or more vanes which are rotated at one or more speeds. The rheometers **130** may measure a torque induced on the vanes by the concrete **101** when the vanes are rotated at the one or more speeds.

[0059] The rheometers **130** may measure the rheological parameters **105** of the concrete **101**. The rheometers **130** may continually measure the rheological parameters **105** of the concrete **101** as the paving machine **100** paves the concrete **101**.

[0060] The controller **108** may receive the rheological parameters **105** of the concrete **101** from the rheometers **130**. The controller **108** may control the actual flow rates **156** of hydraulic fluid to the hydraulic vibrators **118** and/or the actual frequencies **146** of the hydraulic vibrators **118** based on the rheological parameters **105**. For example, the controller **108** may dynamically update the target flow rates **158** and/or the target frequencies **148** based on the rheological parameters **105**. Where the rheological parameters **105** indicate the concrete **101** is easily workable, the target flow rates **158** and/or the target frequencies **148** may be decreased. Where the rheological parameters **105** indicate the concrete **101** is hard to work, the target flow rates **158** and/or the target frequencies **148** may be increased.

[0061] The actual flow rates **156** to the hydraulic vibrators **118** may be continually updated towards the target flow rates **158** and/or the actual frequencies **146** may be continually updated towards the target frequencies **148** based on the actual flow rates **156** measured by the flow meters **154** and/or the actual frequencies **146** measured by the vibration sensors **144** when the target flow rates **158** and/or the target frequencies **148** are dynamically updated based on the rheological parameters **105**. For example, the actual flow rates **156** to the hydraulic vibrators **118** may be continually updated towards the target flow rates **158** based on the actual flow rates **156** measured by the flow meters **154** when the target flow rates **158** are dynamically updated based on the rheological parameters **105**. By way of another example, the actual frequencies **146** may be continually updated towards the target frequencies **148** based on the actual frequencies **146** measured by the vibration sensors **144** when the target frequencies **148** are dynamically updated based on the rheological parameters **105**. Updating the actual flow rates **156** to the hydraulic vibrators **118** towards the target flow rates **158** and/or the actual frequencies **146** towards the target frequencies **148** may enable controlling the actual frequencies **146** of the hydraulic vibrators **118** based on the rheological parameters **105**.

[0062] The controller **108** may include processors **150** and memory **152**. The processors **150** may be configured to execute a set of executable code maintained on the memory **152**. The set of executable code may be configured to cause the processors **150** to carry out the functions of the controller **108**.

[0063] The target frequencies **148** and/or the target flow rates **158** for the rheological parameters **105** may be stored in the memory **152** as a lookup table or similar pre-stored information. The controller **108** may determine the target frequencies **148** and/or the target flow rates **158** by performing a lookup in the lookup table using the rheological parameters **105**. In this regard, the controller **108** may be considered to include software which automatically determines the target frequencies **148** and/or the target flow rates **158**.

[0064] In embodiments, the slipform mold **102** may include multiple of the rheometers **130**. The controller **108** may receive the rheological parameters **105** from the multiple of the rheometers **130**. The controller **108** may perform a weighted average of the rheological parameters **105** when determining the target frequencies **148**. Alternatively, the controller **108** may segment the target frequencies **148** based on a position of the rheometers **130** and the hydraulic vibrators **118**. For

example, the slipform mold **102** may include one of the rheometers **130** on a left-side and one of the rheometers **130** on a right-side of the slipform mold **102**. The controller **108** may determine the target frequencies **148** and/or the target flow rates **158** for the hydraulic vibrators **118** which are on the left-side and the right-side of the slipform mold **102** based on the rheological parameters **105** from the rheometers **130** on the left-side and the right-side, respectively, of the slipform mold **102**. In this regard, the target flow rates **158** and/or the target frequencies **148** may or may not be the same along the width of the slipform mold **102**.

[0065] The paving machine **100** may include the frame **104**. The frame **104** may also be referred to as a framework, a chassis, or the like. The frame **104** may include one or more members. In some embodiments, the frame **104** may include an adjustable width, although this is not intended to be limiting. For example, one or more of the members of the frame may be telescopic (e.g., via one or more roller bearings).

[0066] The paving machine **100** may be a two-track machine, a three-track machine, or a four-track machine. The paving machine **100** may include the end structures **106**. The paving machine **100** may include at least two of the end structures **106**. For example, the paving machine **100** may include two, three, or four (as depicted) of the end structures **106**. The end structures **106** may be coupled to the frame **104**. For example, the end structures **106** may be coupled directly to the frame **104** or coupled to the frame **104** via the pivot arms **116**. The end structures **106** may support at least a portion of the frame **104**. In this regard, the end structures **106** may be configured to support from 10,000 pounds to 27,000 pounds, or more.

[0067] The end structures **106** may include a leg assembly **112**. The leg assembly **112** may be configured to adjust a height of the frame **104** relative to a ground surface. The leg assembly **112** may include an outer tube portion and an inner tube portion coupled by a linear actuator. The outer tube portion and the inner tube portion may be configured to telescope relative to one another. The linear actuator may include a hydraulic cylinder. The hydraulic cylinder may be a smart cylinder including one or more position transducers for determining a height of the leg assembly **112**. The position transducers may include a linear transducer (e.g., a wand and wiper assembly) for monitoring a displacement of the hydraulic cylinder, although this is not intended to be limiting. The leg assembly **112** may include any leg assembly known in the art. For example, the leg assembly **112** may be like a leg assembly described in U.S. Pat. No. 9,764,762, titled ROTARY PIVOT ARM POSITIONING ASSEMBLY, which is incorporated herein by reference in the entirety. By way of another example, the leg assembly **112** may be like a leg assembly described in U.S. Pat. No. 11,254,359, titled LEG ASSEMBLY FOR CONSTRUCTION MACHINE, which is incorporated herein by reference in the entirety.

[0068] The end structures **106** may include a track section **114**. The track section **114** may also be referred to as a crawler assembly, a continuous track, or a caterpillar track, among other names. The track section **114** may be disposed below the leg assembly **112** and coupled to the leg assembly **112**. For example, the track section **114** may be coupled to the leg assembly **112** by a yoke, a slew drive, or the like. Power may be supplied from the hydraulic power supply **110** to the track section **114** (e.g., to a track drive of the track section). In response to receiving the power, the track drive may turn an endless track of the track section **114**. By turning the endless track, the paving machine **100** may be propelled in the paving direction. The track drive may also be used to assist in pivoting the pivot arms **116** (i.e., pivoting on-the-go, stationary pivoting, crab steer, etc.). Thus, the paving machine **100** may be adapted to move in a paving direction by the end structures **106**. The controller **108** may control a speed of the paving machine **100** using the track sections **114**.

[0069] In embodiments, the controller **108** may control the speed of the paving machine **100** based on the rheological parameters **105** measured by the rheometers **130**. For example, the controller **108** may control the track drive of the track section **114** based on the rheological parameters **105**. Where the rheological parameters **105** indicate the concrete **101** is easily workable, the speed of the paving machine **100** may be increased. Where the rheological parameters **105** indicate the concrete

101 is hard to work, the speed of the paving machine **100** may be decreased.

[0070] The paving machine **100** may include pivot arms **116**. The pivot arms **116** may pivotably connect the end structures **106** with the frame **104**. In this regard, each pivot arms **116** may be coupled with an end structures **106**. The pivot arms **116** may be coupled in any manner, such as, but not limited to, a slew drive, a ratcheting assembly, a four-bar-linkage configuration of a hydraulic cylinder, or a planetary drive. Although the paving machine **100** is described as including the pivot arms **116**, this is not intended as a limitation of the present disclosure. For example, the two-track paver machine may or may not include the pivot arms **116**. Furthermore, although the paving machine **100** is depicted as a four-track paving machine, this is not intended as a limitation of the present disclosure. It is further contemplated that the paving machine **100** may be a two-track and/or a three-track paving machine. For example, the paving machine **100** may be a two-track paving machine. The two-track paving machine may include two of the end structures **106** and may not include the pivot arms **116**. By way of another example, the paving machine **100** may be a three-track paving machine. The three-track paving machine may include three of the end structures **106** and one or more of the pivot arms **116**. For example, one or more of the end structures **106** may be coupled to the frame **104** by the pivot arms **116**.

[0071] Referring now to FIGS. 2A-2B, the paving machine **100** is described in accordance with one or more embodiments of the present disclosure. The paving machine **100** may also include depth sensors **202**. The depth sensors **202** may be configured to generate a stream of depth maps **204**. The depth maps **204** may be depth maps from the depth sensors **202** to the concrete **101**.

[0072] The controller **108** may receive the depth maps **204** from the depth sensors **202**. The controller **108** may determine the rheological parameters **105** of the concrete **101** based on the depth maps **204**. For example, the depth to the concrete **101** may vary over time. For instance, the concrete **101** may flow (e.g., slump downwards), thereby changing the depth, where the flow may be based on the rheological parameters. The change in the depth may be used to determine the rheological parameters **105**.

[0073] The depth sensors **202** may be oriented to capture the depth maps **204** of the concrete **101**. For example, the depth sensors **202** may be coupled to one or more of the frame **104**, the pivot arms **116**, the leg assembly **112**, or the track section **114**. The depth sensors **202** may be coupled to the frame **104** and disposed above the hydraulic vibrators **118**. The depth maps **204** may thus include the concrete **101** before consolidation into the slab **103**. In some instances, the depth sensors may have a reduced ability to generate the depth map when covered in particles such as dirt, dust, or concrete. It is contemplated that the depth sensors may be mounted to avoid the particles. For example, the depth sensors **202** may be mounted high enough on the paving machine **100** to avoid the particles with a bird's eye view pointed downwards on the concrete. It is further contemplated that the depth sensors **202** may function even when covered by the particles. For example, the depth sensors **202** may use a wavelength to generate the depth maps **204** which may pass through the particles.

[0074] The depth sensors **202** may include a field-of-view (FOV). The field-of-view of the depth sensors **202** may or may not overlap.

[0075] The depth sensors **202** may be used in combination with or in place of the rheometers **130** to determine the rheological parameters **105**.

[0076] Referring now to FIGS. 3A-3E, the paving machine **100** is described, in accordance with one or more embodiments of the present disclosure. In embodiments, the slab **103** may define surface voids **306**. The surface voids **306** may be defined by a top surface of the slab **103**. The surface voids **306** may also be referred to as surface air voids, pitting, cavities, and the like. A size of the surface voids **306** may be based on the rheological parameters **105** of the concrete **101** and/or the actual frequencies **146** of the hydraulic vibrators **118** at which the slab **103** is consolidated. For example, larger of the surface voids **306** may indicate that the actual frequencies **146** of the hydraulic vibrators **118** at which the slab **103** was consolidated was insufficient.

[0077] In embodiments, the paving machine **100** may include cameras **302**. The cameras **302** may be configured to generate images **304**. The images **304** may be images of the top surface of the slab **103**. The images **304** may include the slab **103** and the surface voids **306**. The images **304** from the cameras **302** may include the slab **103** immediately upon exiting the slipform mold **102**. The cameras **302** may be located at one or more positions to generate the images **304** of the top surface of the slab **103**, such as, but not limited to, suspended above the slab **103**. For example, the cameras **302** may be coupled to the frame **104** above the slipform mold **102**. The cameras **302** may be disposed adjacent to a rear end of the slipform mold **102**. For example, the cameras **302** may be longitudinally disposed between the rear end of the slipform mold **102** and an operator walkway of the paving machine **100**.

[0078] The controller **108** may receive the images **304** from the cameras **302**. The controller **108** may detect the surface voids **306** in the images **304**. For example, the controller **108** may detect the surface voids **306** by detecting a contrast between the surface voids **306** and the slab **103**, by performing one or more image recognition algorithms on the images **304**, or the like. In embodiments, the controller **108** may detect a size, a quantity, or the like of the surface voids **306** in the images **304**.

[0079] In embodiments, the controller **108** may dynamically update the target frequencies **148** and/or the target flow rates **158** based on the surface voids **306** detected in the images **304**. For example, the controller **108** may increase the target flow rates **158** and/or the target frequencies **148** to decrease the size of the surface voids **306** in subsequent of the images **304**. Thus, the surface voids **306** detected in the images **304** may be used to estimate rheological parameters **105** of the concrete **101** after forming the slab **103**. Updating the target flow rates **158** and/or the target frequencies **148** based on the surface voids **306** may be considered reactionary (e.g., after forming), as opposed to methods described above where the target flow rates **158** and/or the target frequencies **148** are dynamically updated based on the rheological parameters **105** determined by the rheometers **130** prior to forming the concrete **101** into the slab **103**.

[0080] Referring now to FIGS. 4A-4B, the paving machine **100** is described, in accordance with one or more embodiments of the present disclosure. Although the paving machine **100** is described as including the hydraulic vibrators **118**, this is not intended as a limitation of the present disclosure. In embodiments, the paving machine **100** may include electric vibrators **402**, electric power source **408**, current meters **410**, vibrator motor controller **412**, and the like. The discussion of the hydraulic vibrators **118** is incorporated herein by reference as to the electric vibrators **402** with appropriate modification from hydraulic to electric. For example, all discussion of controlling the flow rate to the hydraulic vibrators **118** may be modified for the hydraulic vibrators **118** by replacing the flow rate with a current flow to the electric vibrators **402**. In this regard, the various embodiments of controlling the hydraulic vibrators **118** based on the rheological parameters **105**, the depth maps **204**, the images **304**, and the like may similarly apply to controlling the electric vibrators **402** based on the rheological parameters **105**, the depth maps **204**, the images **304**, and the like.

[0081] The electric vibrators **402** may include an electric motor **404** and the eccentric weight **134**. The electric motor **404** may receive electric power from the electric power source **408**. The electric motor **404** may cause the eccentric weight **134** to vibrate in response to receiving the electric power. The electric motor **404** may receive an actual current **406** (I.sub.ACTUAL) at a voltage, thereby defining the electric power. The electric motor **404** may vibrate the eccentric weight **134** at the actual frequencies **146** (f.sub.ACTUAL) in response to receiving the actual current **406**. Increases in the actual current **406** will, in turn, create an increase in the actual frequencies **146**. For example, the actual frequencies **146** may scale linearly with the actual current **406**.

[0082] The paving machine **100** may include current meters **410**. The current meters **410** may be coupled between the vibrator motor controller **412** and the electric vibrators **402**.

[0083] The vibrator motor controller **412** may be coupled between the electric power source **408**

and the electric vibrators **402**. The vibrator motor controller **412** may control the actual current **406** to the electric vibrators **402**. The vibrator motor controller **412** may receive electrical signals from the controller **108**, causing the vibrator motor controller **412** to control the actual current **406** from the electric power source **408** to the electric vibrators **402**. The actual frequencies **146** of the electric vibrators **402** may be controlled by varying the actual current **406** to the electric vibrators **402**. The vibrator motor controller **412** may also control the voltage to the electric motors **404**, thereby controlling the amplitude of the eccentric weights **134**.

[0084] The controller **108** may receive the actual current **406** from the current meters **410** and/or the actual frequencies **146** from the vibration sensors **144**.

[0085] The controller **108** may use the actual current **406** and/or the actual frequencies **146** as feedback when adjusting the actual current **406** to the electric vibrators **402**, thereby moving the actual current **406** towards a target current **414** and/or the actual frequencies **146** towards the target frequencies **148**. The controller **108** may cause the vibrator motor controller **412** to control the actual current **406** to the electric vibrators **402** towards target frequencies **148** using the actual current **406** as feedback.

[0086] In embodiments, the paving machine **100** may include electric vibrators **402-1** through **402-n**. Each of the electric vibrators **402-1** through **402-n** may include a respective of the electric motors **404-1** through **404-n**, the eccentric weights **134-1** through **134-n**, and the like. Similarly, each of the electric vibrators **402-1** through **402-n** may receive a respective of the actual currents **406-1** through **406-n** and may vibrate at respective of the actual frequencies **146-1** through **146-n**. The current meters **410** may include current meters **410-1** through **410-n** coupled between the vibrator motor controller **412** and respective of the electric vibrators **402-1** through **402-n**. The current meters **410-1** through **410-n** may measure the actual current **406-1** through **406-n**.

[0087] Referring now to FIGS. 5A-5B, the paving machine **100** is described, in accordance with one or more embodiments of the present disclosure. In embodiments, the paving machine **100** may include the hydraulic vibrators **118-1** through **118-n** and one or more of the electric vibrators **402**. For example, the electric vibrators **402** may be one vibrator in a bank of the hydraulic vibrators **118**.

[0088] The electric vibrators **402** may be considered a sensor vibrator. In embodiments, the vibrator motor controller **412** may set the electric vibrators **402** at a fixed voltage and a fixed vibrator frequency. The current meter **410** may measure the actual current **406** at the fixed voltage and the fixed vibrator frequency. The actual current **406** may correspond to the rheological parameters **105**. For example, if the concrete **101** is easier to work at the fixed voltage and a fixed vibrator frequency then the actual current **406** may decrease. By way of another example, if the concrete **101** is harder to work at the fixed voltage and a fixed vibrator frequency then the actual current **406** may increase.

[0089] The controller **108** may receive the measurement of the actual current **406** from the current meter **410** and dynamically update the target frequencies **148** and/or the target flow rates **158** of the hydraulic vibrators **118** based on the measurement of the actual current **406** to the electric vibrators **402**.

[0090] Referring now to FIGS. 6A-6D, a system **600** is described, in accordance with one or more embodiments of the present disclosure. The system **600** may include the paving machine **100** and a mixer truck **602**. The mixer truck **602** may place the concrete **101** in front of the paving machine **100**. The paving machine **100** may be communicatively coupled to the mixer truck **602**. The paving machine **100** (e.g., the controller **108**) may receive the rheological parameters **105** of the concrete **101** from the mixer truck **602**, a placement position **604** at which the mixer truck **602** has placed the concrete **101**, and/or a placement time **606** at which the mixer truck **602** has placed the concrete **101**. The mixer truck **602** may determine the rheological parameters **105** based on one or more of a rotational speed of the mixer truck, a mass of the concrete **101** in the mixer truck **602**, a shear rate of the concrete **101**, and the like.

[0091] In embodiments, the controller **108** may receive the rheological parameters **105**, the placement position **604**, and/or the placement time **606**. The controller **108** may then dynamically update the target flow rates **158**, the target frequencies **148**, the target current **414**, and the like based on the rheological parameters **105**, the placement position **604**, and/or the placement time **606**. For example, the controller **108** may account for changes in the rheological parameters **105** between batches of the concrete **101** (e.g., batch of concrete **101-1**, **101-2**) based on the placement position **604** and the current position of the paving machine **100**. By way of another example, the controller **108** may account for changes in the rheological parameters **105** due to time elapsing since the placement time **606**. For instance, the water content of the concrete **101** may decrease (e.g., causing slump loss) based on time exposed on the ground before being formed. Thus, the controller **108** may dynamically update the target flow rates **158**, the target frequencies **148**, the target current **414**, and the like based on the information received from the mixer truck to further include a reference to position and time of day for where and when the concrete was placed, since the rheological parameters **105** may change over time.

[0092] One or more meteorological parameters may change the rate at which the water content of the concrete **101** decreases between the placement time **606** and the time at which the paving machine **100** consolidates the concrete **101** into the slab **103**. The meteorological parameters may include ambient temperature **608**, wind speed **610**, humidity **612** (e.g., absolute humidity, relative humidity, or specific humidity), air pressure **614**, and the like. The rate at which the water content of the concrete **101** decreases may be proportional to the ambient temperature **608**, proportional to the wind speed **610**, inversely proportional to the humidity **612**, and/or inversely proportional to the air pressure **614**. With an increase in the ambient temperature **608**, an increase in the wind speed **610**, a decrease in the humidity **612**, and/or a decrease in the air pressure **614**, the concrete **101** may experience higher loss of water content due to evaporation such that the workability may decrease as compared to concrete **101** which is exposed to a lower ambient temperature, a lower wind speed, a higher humidity, and/or a higher air pressure over the same period. In this regard, the rate of change of the rheological parameters **105** of the concrete **101** over time after being placed by the mixer truck **602** in front of the paving machine **100** may be dependent upon the ambient temperature **608**, the wind speed **610**, the humidity **612**, and/or the air pressure **614**.

[0093] In embodiments, the controller **108** may receive the meteorological parameters. For example, the controller **108** may receive the ambient temperature **608**, wind speed **610**, humidity **612**, air pressure **614**, and the like. The paving machine **100** may include one or more meteorological sensors by which the controller **108** may receive the meteorological parameters, such as, but not limited to, an ambient temperature sensor (e.g., thermometer **609**), a wind speed sensor and/or wind direction sensor (e.g., anemometer **611**), a humidity sensor (e.g., hygrometer **613**), an air pressure sensor (e.g., barometer **615**), and the like. The thermometer **609**, the hygrometer **613**, and the barometer **615** may measure the ambient temperature **608**, wind speed **610**, humidity **612**, and air pressure **614**, respectively.

[0094] The controller **108** may dynamically update the rheological parameters **105** received from the mixer truck **602** based on the placement time **606** in combination with the ambient temperature **608**, wind speed **610**, humidity **612**, and/or the air pressure **614**. The controller **108** may then control the target flow rates **158**, the target frequencies **148**, the target current **414**, and the like based on the rheological parameters **105** which has been updated. Thus, the controller **108** may compensate for the change in the water content when controlling the hydraulic vibrators **118** and/or the electric vibrators **402**.

[0095] Although not depicted, the system **600** may further include one or more placer/spreaders to place and/or spreader the concrete **101** from the mixer truck **602** in front of the paving machine **100**.

[0096] FIG. 7 depicts the paving machine **100**, in accordance with one or more embodiments of the present disclosure. In this example, the paving machine **100** is configured as a two-track machine

with two of the end structures **106**. The paving machine is also configured with the split front auger **126**, although this is not intended to be limiting. The two-track machine may be configured with any of the various embodiments, such as the rheometers **130**, the depth sensors **202**, the cameras **302**, the sensor vibrators, and the like.

[0097] FIGS. **8A-8C** depict the paving machine **100**, in accordance with one or more embodiments of the present disclosure. Although much of the present disclosure has described the paving machine **100** as a slipform paver with the slipform mold **102** configured to form the slab **103**, this is not intended as a limitation of the present disclosure. The paving machine **100** may be a curb-and-gutter paving machine (see FIG. **8A**, configured as a three-track machine), a barrier paving machine (see FIG. **8B**, configured as a four-track machine), a sidewalk paving machine, or the like. The slipform mold **102** may be mounted to a side of the frame **104**. The slipform mold **102** may be an offset mold mounted to the side of the frame **104**. The slipform mold **102** may be mounted in either a right-hand pour or left-hand pour configuration. The slipform mold **102** may be configured to form the concrete **101** into a curb-and-gutter **802**, a barrier **804**, or the like as the paving machine **100** travels in the paving direction. The slipform mold **102** may form the curb-and-gutter **802** and/or the barrier **804** in a continuous operation.

[0098] The slipform mold **102** include a hopper **806**. The hopper **806** may also be referred to as a chute. The slipform mold **102** may also include a forming section **808**. The forming section **808** may be disposed behind the hopper **806** in the paving direction. The forming section **808** may receive the concrete **101** from the hopper **806**. The forming section **808** may form the curb-and-gutter **802** and/or the barrier **804**. The forming section **808** may include any suitable profile for forming the material into the curb-and-gutter **802**, such as, but not limited to, straight curbs, mower curbs, or slanted curbs. Similarly, the forming section **808** may include any suitable profile for forming the material into the barrier **804**, such as, but not limited to, a median barrier, parapet barrier, or the like.

[0099] The hydraulic vibrators **118** may be immersion vibrators which may be affixed within the hopper **806**. The hydraulic vibrators **118** may be disposed a select distance away from the walls of the hopper **806**. The hydraulic vibrators **118** may be disposed at a height aligned with the concrete **101** in the forming section **808**.

[0100] The paving machine **100** with the slipform mold **102** in the offset configuration may include any of the various embodiments for determining the rheological parameters **105**, such as the rheometers **130**, the cameras **302**, the sensor vibrators, and the like. For example, the rheometers **130** may be disposed within the hopper **806**. The hopper **806** may constrain the concrete **101**, enabling the rheometers **130** to measure the rheological parameters **105** as the concrete **101** falls down the hopper **806** into the forming section **808**. By way of another example, the cameras **302** may be coupled to a side of the frame **104** behind the slipform mold **102** in the paving direction for generating the images **304**. By way of another example, the electric vibrators **402** may be disposed within the hopper **806** for sensing the rheological parameters **105**. The electric vibrators **402** may replace one or more of the hydraulic vibrators **118** in existing of the slipform molds **102**.

[0101] It is contemplated that the paving machine **100** with the slipform mold **102** in the offset configuration may or may not include the depth sensors **202**. For example, the concrete **101** is not poured ahead of the slipform mold **102**, such that the depth sensors **202** may not function in the offset configuration.

[0102] Referring generally again to the figures. As may be understood, the specific components of the hydraulic circuit, sub-circuits, and the paths to/from the hydraulic sub-circuits of the paving machine **100** are not provided for clarity. It is further contemplated that the paving machine **100** may include hydraulic components, such as, but not limited to, inlet manifolds, outlet manifolds, hydraulic motors, hydraulic cylinders, filters, check valves, relief valves, and the like.

[0103] The slipform mold **102** may or may not include an adjustable width. The slipform mold **102** may be a fixed width mold, an adjustable width, or the like. For example, the slipform mold **102**

may be an adjustable width paving mold. The width between the side plates **124** may be adjustable. The adjustable width paving mold may provide the paving machine **100** with an ability to form tapered slabs on-the-go by a paving width change. The adjustable width paving mold may include a hydraulically telescoping rolling frame with dual rollers.

[0104] In the case of a control algorithm, one or more program instructions or methods may be configured to operate via proportional control, feedback control, feedforward control, integral control, proportional-derivative (PD) control, proportional-integral (PI) control, proportional-integral-derivative (PID) control, or the like.

[0105] The controller **108** may be communicatively coupled with one or more electronic components of the paving machine **100**. For example, the controller **108** may be communicatively coupled with the hydraulic vibrators **118** (e.g., vibration sensors **144**), the hydraulic manifold **140** (e.g., adjustable flow control valves **142**), and/or the rheometers **130**. The controller **108** may be communicatively with the electronic components by one or more controller area network buses or the like.

[0106] The controller may be a mobile machine control computer, a desktop computer, workstation, parallel processor, or other computer system configured to execute a program configured to operate the paving machine **100**, as described throughout the present disclosure.

[0107] As may be understood, the processors may include any one or more processing elements known in the art. In this sense, the processors may include any microprocessor-type device configured to execute software algorithms and/or instructions. For the purposes of the present disclosure, the term “processor” or “processing element” may be broadly defined to encompass any device having one or more processing or logic elements (e.g., one or more micro-processor devices, one or more application specific integrated circuit (ASIC) devices, one or more field programmable gate arrays (FPGAs), or one or more digital signal processors (DSPs)). In this sense, the one or more processors may include any device configured to execute algorithms and/or instructions (e.g., program instructions stored in memory).

[0108] Furthermore, the memory may include any storage medium known in the art suitable for storing program instructions executable by the associated one or more processors. For example, the memory medium may include a non-transitory memory medium. By way of another example, the memory medium may include, but is not limited to, a read-only memory (ROM), a random-access memory (RAM), a magnetic or optical memory device (e.g., disk), a solid-state drive and the like. It is further noted that memory medium may be housed in a common controller housing with the one or more processors. In one embodiment, the memory medium may be located remotely with respect to the physical location of the one or more processors.

[0109] All of the methods described herein may include storing results of one or more steps of the method embodiments in memory. The results may include any of the results described herein and may be stored in any manner known in the art. The memory may include any memory described herein or any other suitable storage medium known in the art. After the results have been stored, the results can be accessed in the memory and used by any of the method or system embodiments described herein, formatted for display to a user, used by another software module, method, or system, and the like. Furthermore, the results may be stored “permanently,” “semi-permanently,” temporarily,” or for some period of time. For example, the memory may be random access memory (RAM), and the results may not necessarily persist indefinitely in the memory. It is further contemplated that each of the embodiments of the method described above may include any other step(s) of any other method(s) described herein. In addition, each of the embodiments of the method described above may be performed by any of the systems described herein.

[0110] One skilled in the art will recognize that the herein described components operations, devices, objects, and the discussion accompanying them are used as examples for the sake of conceptual clarity and that various configuration modifications are contemplated. Consequently, as used herein, the specific exemplars set forth and the accompanying discussion are intended to be

representative of their more general classes. In general, use of any specific exemplar is intended to be representative of its class, and the non-inclusion of specific components, operations, devices, and objects should not be taken as limiting.

[0111] As used herein, directional terms such as “top,” “bottom,” “front,” “back,” “over,” “under,” “upper,” “upward,” “lower,” “down,” and “downward” are intended to provide relative positions for purposes of description, and are not intended to designate an absolute frame of reference.

Various modifications to the described embodiments will be apparent to those with skill in the art, and the general principles defined herein may be applied to other embodiments.

[0112] With respect to the use of substantially any plural and/or singular terms herein, those having skill in the art can translate from the plural to the singular and/or from the singular to the plural as is appropriate to the context and/or application. The various singular/plural permutations are not expressly set forth herein for sake of clarity.

[0113] It is believed that the present disclosure and many of its attendant advantages will be understood by the foregoing description, and it will be apparent that various changes may be made in the form, construction and arrangement of the components without departing from the disclosed subject matter or without sacrificing all of its material advantages. The form described is merely explanatory, and it is the intention of the following claims to encompass and include such changes. Furthermore, it is to be understood that the invention is defined by the appended claims.

Claims

1. A paving machine comprising: a slipform mold, wherein the slipform mold is configured to form concrete, wherein the slipform mold comprises: a plurality of hydraulic vibrators, wherein the plurality of hydraulic vibrators are configured to vibrate in response to receiving hydraulic fluid; and one or more rheometers, wherein the one or more rheometers are configured to measure rheological parameters of the concrete; a hydraulic power supply comprising a hydraulic pump; a hydraulic manifold, wherein the hydraulic manifold is coupled between the hydraulic pump and the plurality of hydraulic vibrators; and a controller comprising one or more processors configured, via executable code, to: cause the hydraulic manifold to control actual flow rates of the hydraulic fluid from the hydraulic pump to the plurality of hydraulic vibrators thereby controlling actual frequencies of the plurality of hydraulic vibrators; and dynamically update at least one of target flow rates of the hydraulic fluid to the plurality of hydraulic vibrators or target frequencies of the plurality of hydraulic vibrators based on the rheological parameters.

2. The paving machine of claim 1, wherein the plurality of hydraulic vibrators comprise a plurality of vibration sensors, wherein the plurality of vibration sensors are configured to measure the actual frequencies of the plurality of hydraulic vibrators, wherein the controller is configured to use the actual frequencies as feedback when causing the hydraulic manifold to control the actual flow rates of the hydraulic fluid from the hydraulic pump to the plurality of hydraulic vibrators thereby controlling the actual frequencies of the plurality of hydraulic vibrators, wherein the actual frequencies are continuously updated towards the target frequencies based on the actual frequencies measured by the plurality of vibration sensors.

3. The paving machine of claim 1, wherein the paving machine comprises a plurality of flow meters, wherein the plurality of flow meters are configured to measure the actual flow rates, wherein the controller is configured to use the actual flow rates as feedback when causing the hydraulic manifold to control the actual flow rates of the hydraulic fluid from the hydraulic pump to the plurality of hydraulic vibrators thereby controlling the actual frequencies of the plurality of hydraulic vibrators, wherein the actual flow rates are continuously updated towards the target flow rates based on the actual flow rates measured by the plurality of flow meters.

4. The paving machine of claim 3, wherein the plurality of flow meters are coupled between the hydraulic manifold and the plurality of hydraulic vibrators.

5. The paving machine of claim 1, wherein the one or more rheometers are vane rheometers.
6. The paving machine of claim 1, wherein the one or more rheometers comprise a plurality of rheometers, wherein the controller is configured to perform a weighted average of the rheological parameters from the plurality of rheometers.
7. The paving machine of claim 1, wherein the controller is configured to control a speed of the paving machine based on the rheological parameters.
8. The paving machine of claim 1, wherein the paving machine comprises one or more cameras, wherein the one or more cameras are configured to generate images of a top surface of the concrete, wherein the controller is configured to receive the images, detect one or more surface voids in the images, and dynamically update at least one of the target frequencies or the target flow rates based on the one or more surface voids detected in the images.
9. The paving machine of claim 1, wherein the paving machine comprises an electric vibrator, a vibrator motor controller, and a current meter, wherein the vibrator motor controller is configured to set the electric vibrator to a fixed voltage and fixed vibrator frequency, wherein the current meter is configured to measure an actual current to the electric vibrator, wherein the controller is configured to dynamically update at least one of the target frequencies or the target flow rates based on the actual current.
10. The paving machine of claim 1, wherein the slipform mold is an inset mold and is configured to form the concrete into a slab, wherein the slipform mold comprises a finishing pan and a pair of side plates, wherein the one or more rheometers are laterally disposed between the pair of side plates.
11. The paving machine of claim 10, wherein the one or more rheometers are longitudinally disposed between the plurality of hydraulic vibrators and the finishing pan.
12. The paving machine of claim 11, wherein the slipform mold comprises a grout box auger, wherein the grout box auger spans between the pair of side plates, wherein the grout box auger is longitudinally disposed between the plurality of hydraulic vibrators and the finishing pan.
13. The paving machine of claim 12, wherein a rotational speed of the grout box auger is adjusted based on at least one of a speed of the paving machine or the rheological parameters.
14. The paving machine of claim 11, wherein the one or more rheometers comprise a plurality of rheometers, wherein the controller is configured to dynamically update at least one of the target frequencies or the target flow rates for the plurality of hydraulic vibrators which are on a left-side and a right-side of the slipform mold based on the rheological parameters from the plurality of rheometers which are disposed on respective of a left-side and a right-side, respectively, of the slipform mold.
15. The paving machine of claim 11, comprising one or more depth sensors, wherein the one or more depth sensors are configured to generate depth maps of the concrete, wherein the controller is configured to receive the depth maps and determine the rheological parameters based on the depth maps in combination with the one or more rheometers.
16. The paving machine of claim 15, wherein the one or more depth sensors are disposed above the plurality of hydraulic vibrators.
17. The paving machine of claim 1, wherein the slipform mold is an offset mold and is configured to form the concrete into one of a curb-and-gutter or a barrier, wherein the slipform mold comprises a hopper and a forming section, wherein the one or more rheometers are disposed in the hopper.
18. The paving machine of claim 1, wherein the paving machine is one of a two-track machine, a three-track machine, or a four-track machine.
19. The paving machine of claim 1, wherein the paving machine is communicatively coupled with a mixer truck, wherein the paving machine is configured to receive the rheological parameters, a placement position, and a placement time from the mixer truck, wherein the controller is configured to dynamically update at least one of the target flow rates or the target frequencies based on the rheological parameters, the placement position, and the placement time.

20. A paving machine comprising: a slipform mold, wherein the slipform mold is configured to form concrete, the slipform mold comprising: a plurality of electric vibrators, wherein the plurality of electric vibrators are configured to vibrate in response to receiving electric power; and one or more rheometers, wherein the one or more rheometers are configured to measure rheological parameters of the concrete; an electric power source; a vibrator motor controller, wherein the vibrator motor controller is coupled between the electric power source and the plurality of electric vibrators; and a controller comprising one or more processors configured, via executable code, to: cause the vibrator motor controller to control actual current from the electric power source to the plurality of electric vibrators thereby controlling actual frequencies of the plurality of electric vibrators; and dynamically update at least one of target current to the plurality of electric vibrators or target frequencies of the plurality of electric vibrators based on the rheological parameters.
